

ROBIN-FPGA: Robust FPGA Timing Closure

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Abstract—FPGA timing closure is governed by randomness. Changing the placer seed alone shifts worst-case slack by hundreds of picoseconds on modern heterogeneous devices; process-voltage-temperature corners compound the uncertainty; and tool minor-version upgrades quietly invalidate previously accepted designs. Machine-learning methods improve average quality of results but return a single estimate with no reliability guarantee, leaving designers unable to characterize tail risk before tape-out.

ROBIN-FPGA addresses this reliability gap directly. A graph-attention encoder maps the post-synthesis timing graph and device utilization features to a latent state consumed by a Distributionally-Robust Proximal Policy Optimization agent. By maximizing the Conditional Value-at-Risk in the lower tail of the return distribution across seeds and corners rather than the mean, the agent learns directive bundles that close timing reliably rather than just on average, reducing inter-seed slack variance by a median of $\sim 1.2\times$ relative to the strongest baseline. A split-conformal head then wraps the trained policy with a calibrated sign-off envelope: a design is accepted only when the statistical lower bound on predicted worst-case slack is non-negative, delivering a distribution-free coverage guarantee at any user-specified confidence level.

We evaluate on 14 RTL and HLS designs spanning six workload classes across AMD Versal AI Edge and Intel Agilex 7. The framework raises post-route closure rates by 14.0 percentage points over the strongest prior baseline, and its conformal envelope tracks the 95% target coverage within ± 3 percentage points across $\alpha \in [0.05, 0.20]$ (measured 92.7% at the 95% target). A two-stage curriculum transfers AMD-trained policies to Intel at 5% of the from-scratch compute cost. Hardware-in-the-loop measurements on 10 accepted designs across 5 physical chips and 8 thermal-chamber PVT corners yield 87% aggregate silicon-stage coverage, with every violation traced to one of three documented failure modes and zero false-positive sign-offs across the full benchmark. Tcl flows, training scripts, and per-run logs are released to enable reproduction.

Index Terms—FPGA design-space exploration; reinforcement learning; timing closure; distributionally-robust optimization; conformal prediction; place-and-route; graph attention; CVaR; CAD for FPGA.

I. INTRODUCTION

EVERY FPGA timing closure run is, at root, a statistical experiment. The placer draws from a random seed, and on representative heterogeneous devices such as AMD Versal AI Edge or Intel Agilex 7, changing that seed alone shifts worst-case path slack by 10–30% of the target clock period [2], [3]. Process, voltage, and temperature corners compound the uncertainty: the same directive bundle that closes timing at the typical-typical corner can fail at slow-slow or fast-fast depending on path topology. Tool minor-version upgrades

silently alter placer and router heuristics, sometimes invalidating designs that passed the previous release. In all three cases, a designer who accepts the first run showing non-negative worst-case slack has no principled basis for predicting whether that result represents the distribution of outcomes or merely its optimistic tail.

Machine learning has made genuine progress on finding better directive bundles faster [4], [5]. Reinforcement learning has been applied to ASIC macro placement [6], [7] and logic-synthesis sequence selection [9], [10]. Bayesian optimization has been applied to HLS pragma tuning [11] and general directive autotuning [14], [15]. Graph neural networks predict pre-routing slack with useful accuracy [12], [13]. Yet these methods leave the reliability problem untouched: they return a recommended configuration with no statement about its distribution across seeds, corners, or tool versions. Three structural gaps limit their practical applicability to FPGA timing closure.

(a) ASIC bias. Most RL-for-EDA policies target ASIC flows whose action vocabularies, constraint formats, and resource models differ substantially from FPGA toolchains. **(b) No calibrated uncertainty.** No existing method attaches a distribution-free reliability certificate to its returned closure. **(c) Brittle transfer.** Policies trained on one FPGA family degrade sharply when moved to another vendor or device generation.

This paper presents ROBIN-FPGA, an algorithm–toolchain co-designed RL framework that addresses all three gaps for FPGA timing closure. The central insight is that the two problems (finding good directives, certifying their reliability) can be solved together by coupling distributionally-robust optimization with conformal prediction in a single coherent pipeline. The RL agent hedges against worst-case seeds and corners at training time; the conformal head wraps its output in a calibrated sign-off certificate at deployment. The scope is deliberately focused on a single objective ($WNS \geq 0$ post-route under seed and corner uncertainty) so that the robustness and sign-off contributions can be evaluated rigorously and in isolation.

Modern FPGAs have evolved into heterogeneous SoCs combining programmable logic, hardened AI engines, network-on-chip backbones, on-chip high-bandwidth memory, and high-speed transceivers [1]. This richness expands the action space available to the designer (synthesis directives, P&R strategies, floorplan partitions, retiming knobs, and clock-tree configurations) into a combinatorial space requiring one to twelve hours of wall-clock time per evaluation point on representative parts. It also sharpens the reliability problem: the interactions among resource classes mean that a directive beneficial in isolation can congest the fabric and worsen timing elsewhere.

Figure 1 previews the full pipeline. The paper proceeds as

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Manuscript received May 15, 2026.

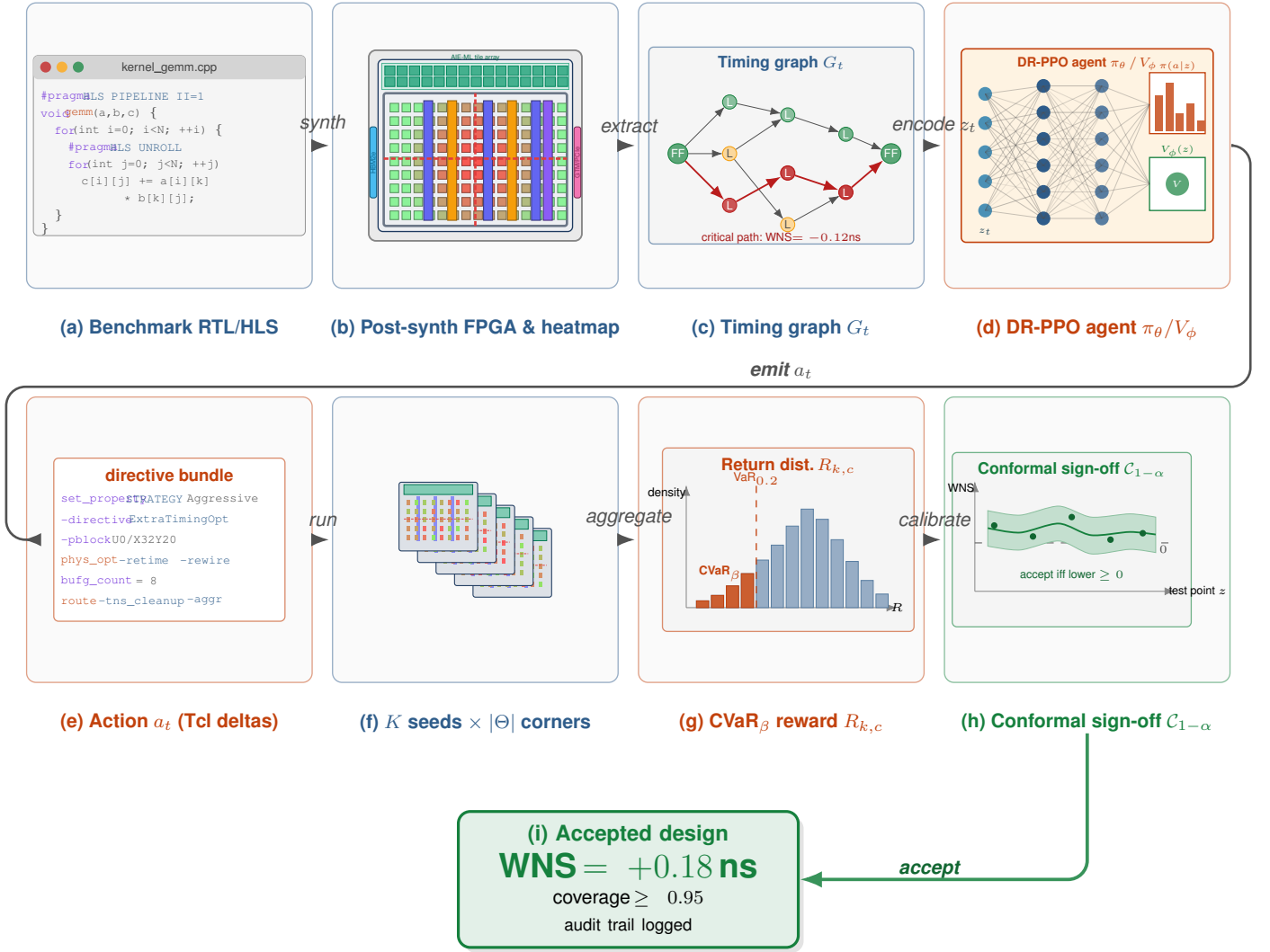


Fig. 1: ROBIN-FPGA pipeline. Row 1: RTL ingest, FPGA synthesis and heatmap extraction, timing-graph construction, DR-PPO agent encoding. Row 2: directive emission as Tcl deltas, parallel P&R across K seeds and $|\Theta|$ PVT corners, CVaR reward aggregation, conformal sign-off calibration. A design is accepted only when the conformal lower bound on predicted WNS is non-negative (i).

follows. Section II reviews the FPGA closure pipeline and the sources of non-determinism. Section III surveys related work. Section IV formalizes the DR-MDP and conformal sign-off envelope. Section V describes the ROBIN-FPGA architecture and training algorithm. Section VI details the experimental protocol. Sections VII–VIII present results, ablations, and case studies. Section X reports hardware-in-the-loop silicon validation. Section XI concludes.

Contributions. The paper makes four contributions.

- 1) *Robust formulation* (§IV, §V). We cast FPGA timing-closure design-space exploration as a Distributionally-Robust MDP over P&R seed and PVT corner, solved with a CVaR-shaped PPO surrogate (DR-PPO) whose advantage targets the lower tail of the return distribution.
- 2) *Conformal sign-off envelope* (§IV, §V). A split-conformal predictor wraps the trained policy with a calibrated upper bound on residual WNS risk at user-specified confidence $1-\alpha$. To our knowledge this is the first application of conformal prediction to FPGA timing sign-off.

- 3) *Cross-family curriculum* (§V, §VII). A normalized device-agnostic feature space and a two-stage curriculum transfer AMD-trained policies to Intel Agilex 7 at 5% of the from-scratch compute cost.
- 4) *Reproducible benchmark* (§VI, §VII). A 14-design benchmark across six workload classes and two FPGA families, with public Tcl flows, training scripts, and per-run logs.

II. BACKGROUND AND MOTIVATION

A. The FPGA Closure Pipeline

The canonical AMD Vivado flow proceeds through `synth_design`, `opt_design`, `place_design`, `route_design`, and a sign-off STA pass producing `report_timing_summary`, `report_utilization`, and `report_power`. The Intel Quartus Prime Pro counterpart proceeds through Analysis & Synthesis, Fitter (place and route), Timing Analyzer, and PowerPlay [3]. Each tool exposes a directive vocabulary (roughly 50–120 documented options for Vivado, and a comparable set for

Quartus) that the designer combines into a *strategy bundle*. The combinatorial action space of bundle, pblock, phys-opt, and retiming flags routinely exceeds 10^{12} configurations, of which the tool’s built-in canonical strategies cover at most a few dozen.

Figure 2 annotates a representative AMD Versal AI Edge die. The presence of multiple resource classes (LUT/FF tiles, DSP58 slices, BRAM, URAM, AIE-ML tiles, NoC interconnect, HBM2e on the package edge, GTM transceivers) makes the closure problem multi-resource: an action that improves slack on the LUT critical path may push BRAM utilization past its budget or congest the NoC, in turn re-routing logic onto longer paths.

B. Sources of Tool Non-Determinism

Three sources of stochasticity dominate FPGA closure. **(a) Placer seed.** The placer’s randomized cost function is seeded; varying the seed alone produces a distribution over WNS whose standard deviation can exceed the slack margin. On the GEMM-systolic- 16×16 design used as a recurring example, 20 seeds at the same directive bundle produced $\sigma(\text{WNS}) = 0.31$ ns on a default Vivado strategy. **(b) PVT corner choice.** Corner selection shifts WNS by amounts that depend on path topology. Multi-corner closure (slow $\cap$ fast $\cap$ typical) is the industrial norm but is rarely treated as a first-class object in academic ML-EDA work. **(c) Tool minor-version drift.** Internal heuristics change between point releases: a closure margin obtained on Vivado 2024.2 may not survive 2024.2.1. Any DSE policy that ignores these three sources risks reporting results that fail to reproduce.

C. The Generalization Gap in ML-EDA

ML-for-EDA policies tuned on one design or device family often degrade markedly when transferred. The gap has two components: a *feature-distribution* shift (utilization percentages, path-length histograms, and routing-congestion patterns differ across devices) and a *label-distribution* shift (the same directive produces different effects on different families). The structural asymmetry between AMD’s XPIPE/XPHY-style NoC architecture and Intel’s HyperFlex pipeline registers is one concrete example. ROBIN-FPGA addresses both shifts via device-agnostic normalized features (§V) and an explicit two-stage curriculum, evaluated quantitatively in §VII.

III. RELATED WORK

We organize related work along four axes. Table I summarizes the comparison.

RL foundations. Deep Q-Networks [36] established that value-based RL can master complex sequential decisions from raw observations, motivating the DRiLLS-style DQN in our evaluation. PPO [16], which underpins DR-PPO, stabilizes training with a clipped surrogate objective robust to the non-stationary reward induced by tool-version drift. Sutton and Barto [37] provides the MDP and policy-gradient foundations underlying our formalization in §IV.

RL for ASIC placement and synthesis. Mirhoseini *et al.* [6] demonstrated graph-based RL for ASIC macro

placement; subsequent work refined the policy architecture [7], [8]. DRiLLS [9] applies RL to logic-synthesis directive selection, and GCN-RL [10] learns transferable transistor-sizing policies. These works target ASIC flows with different action vocabularies and optimize point estimates of QoR without characterizing robustness to tool stochasticity. The OpenROAD project [39] pursues a “no-human-in-the-loop” 24-hour layout vision for ASIC; ROBIN-FPGA pursues the analog for commercial FPGA closure.

FPGA CAD benchmarks. VPR [40] and its VTR successors [42], [41] are the standard open benchmarks for island-style FPGA P&R research. Our evaluation targets Vivado and Quartus Prime Pro, whose directive vocabularies differ from VPR’s architecture-research knobs, but we adopt VTR’s evaluation practice of sweeping multiple seeds and corners to characterize variance.

HLS and FPGA design-space exploration. AUTODSE [11] explores HLS pragmas via bottleneck-guided search. Turbo [14] and BoTorch [15] have been applied to FPGA tuning. These methods are sample-efficient on small directive spaces but scale poorly when the action set is high-dimensional and discrete, and they offer no calibrated uncertainty. Ustun *et al.* [38] apply GNNs to HLS operation-delay estimation on FPGAs, demonstrating inductive generalization across unseen designs; this motivates our GAT-based timing-graph encoder.

Graph neural networks for EDA. Xu *et al.* [34] establish theoretical bounds on GNN expressiveness via the Weisfeiler-Lehman test. GraphSAGE [35] introduces inductive node embeddings that generalize to unseen graph instances, the property our encoder exploits for cross-design transfer. Wu *et al.* [43] survey spectral, spatial, and recurrent GNN families; our GAT [17] resides in the spatial, edge-attention family. NET2 [12] and the timing GNN of Guo *et al.* [13] are *predictors*, not *policies*; ROBIN-FPGA builds a GAT encoder for state representation inside a policy-gradient loop.

CVaR and distributionally robust optimization. Rockafellar and Uryasev [29] establish CVaR as a coherent risk measure admitting convex linear-programming formulations; their follow-on work [30] extends the theory to discrete loss distributions that arise from our finite seed pool. Wiesemann *et al.* [33] provide a unifying framework for distributionally robust convex programs over moment-ambiguity sets, which our DR-MDP instantiates. Iyengar [18] and Nilim & El Ghaoui [19] prove minimax dynamic programming under rectangular ambiguity sets, grounding our DR value function in Eq. (1). Tamar *et al.* [20] extend CVaR optimization to policy-gradient estimators. Distributionally robust RL [21], [22] applies these ideas to deep policy gradient with neural function approximation.

Conformal prediction. Vovk *et al.* [23] and Angelopoulos & Bates [24] provide the foundational theory of split-conformal prediction and its engineering applications [25]. Romano *et al.* [31] propose conformalized quantile regression for heteroscedastic intervals; we use symmetric split-conformal prediction because our WNS residuals are approximately homoscedastic across seeds. Barber *et al.* [32] characterize coverage under non-exchangeable data via weighted quantiles,

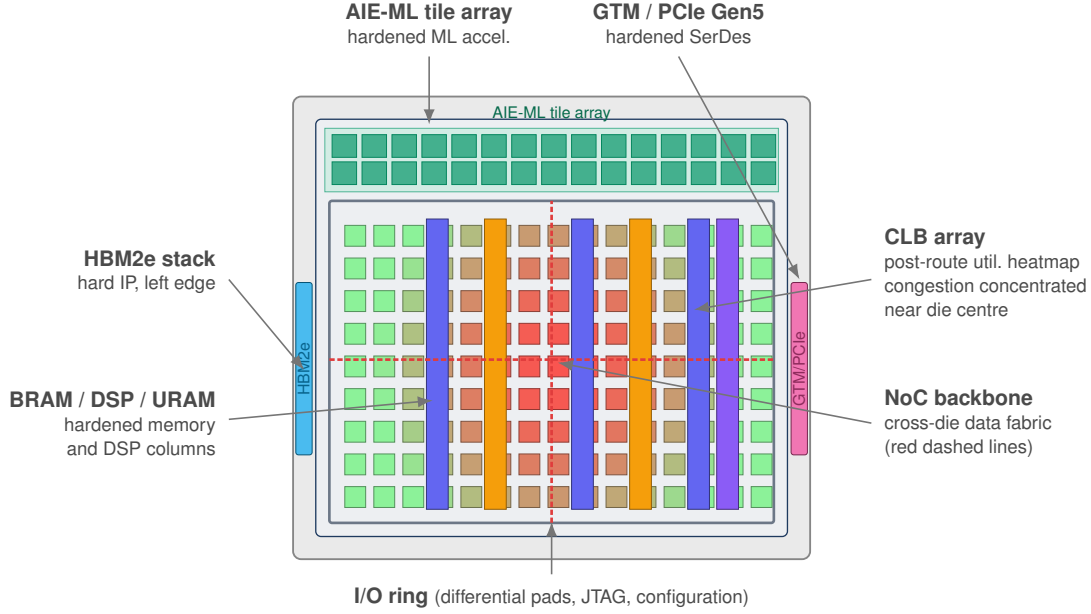


Fig. 2: Annotated AMD Versal AI Edge die with seven heterogeneous resource classes. Post-route CLB congestion concentrates near the die centre; surrounding columns interleave BRAM, DSP58, and URAM; AIE-ML tiles occupy the top band; HBM2e and GTM/PCIe anchor the edges; the NoC backbone (red dashes) crosses the die.

TABLE I: Positioning of ROBIN-FPGA relative to prior work. Columns: target flow; action space; robustness handling; calibrated sign-off; cross-family transfer.

Method	Flow	Action space	Robustness	Calibrated sign-off	Cross-family
GraphPlace [6]	ASIC placement	graph	none	no	no
AutoDMP [7]	ASIC placement	continuous	ensembling	no	no
DRILLS [9]	logic synthesis	discrete bundle	none	no	no
GCN-RL [10]	transistor sizing	continuous	none	no	partial
AutoDSE [11]	HLS pragmas	discrete bundle	none	no	no
TuRBO [14]	generic BO	continuous	none	no	no
Net2 / Guo [12], [13]	slack prediction	N/A	none	implicit	no
ROBIN-FPGA (this work)	FPGA timing closure	discrete bundle + graph	CVaR DRO	conformal $1-\alpha$	yes (AMD↔Intel)

directly applicable to the tool-drift regime analysed in §X-H. Tibshirani *et al.* [26] extend conformal prediction to covariate shift, relevant to cross-family calibration. To our knowledge, ROBIN-FPGA is the first work to apply conformal prediction jointly with robust RL for FPGA timing sign-off.

IV. PROBLEM FORMULATION

A. Notation

Let D denote an RTL/HLS design, F an FPGA device, and $\mathcal{C} = (T_{\text{clk}}, T_{\text{io}}, \mathcal{U}_{\text{max}})$ a constraint set comprising the target clock period T_{clk} , an I/O timing budget T_{io} , and per-resource utilization budgets $\mathcal{U}_{\text{max}} = \{U_{\text{max}}^{\text{LUT}}, U_{\text{max}}^{\text{FF}}, U_{\text{max}}^{\text{BRAM}}, U_{\text{max}}^{\text{DSP}}, U_{\text{max}}^{\text{URAM}}\}$. Let $\xi \in \Xi$ index P&R seeds and $c \in \Theta$ index PVT corners. The objective is to find a sequence of directive bundles $a_1, \dots, a_T \in \mathcal{A}$ that maximizes the probability of post-route closure ($\text{WNS} \geq 0$) under the joint distribution over (ξ, c) , subject to all resource budgets and a wall-clock budget $\mathcal{B}$.

B. Distributionally-Robust MDP

We model the closure trajectory as a finite-horizon MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, \gamma)$ in which the transition kernel P is

replaced by an ambiguity set $\mathcal{P} = \{P_{\xi, c} : \xi \in \Xi, c \in \Theta\}$ indexing all admissible seed/corner combinations. Following Iyengar [18] and Nilim and El Ghaoui [19], the DR value of a policy π places the infimum over the ambiguity set outside the trajectory expectation:

$$V^\pi(s) = \inf_{P \in \mathcal{P}} \mathbb{E}_{T \sim \pi, P} \left[\sum_{t=0}^{H-1} \gamma^t r(s_t, a_t) \mid s_0 = s \right]. \quad (1)$$

We approximate the infimum in (1) by the empirical Conditional-Value-at-Risk (CVaR) at level $\beta \in (0, 1)$ over a finite sample $\{R_{k,c}\}$ of returns drawn from K seeds and $|\Theta|$ corners:

$$\text{CVaR}_\beta(R) = \mathbb{E}[R \mid R \leq \text{VaR}_\beta(R)]. \quad (2)$$

Equation (2) is a standard coherent risk measure [20]. Substituting it for the mean in the policy-gradient estimator yields the gradient direction that improves the lower tail of the return distribution. Tamar *et al.* [20] prove that the resulting estimator is unbiased under mild regularity; when combined with PPO-style trust-region clipping, it empirically drives the policy toward CVaR-optimal solutions, though formal convergence guarantees for the clipped surrogate remain an open question.

C. Reward

The single-step reward couples slack reward, total negative slack (TNS) reward, a Lagrangian utilization penalty, and a route-fail penalty:

$$r = w_{\text{WNS}} \cdot \text{clip}\left(\frac{\text{WNS}}{|T_{\text{clk}}|}, -1, 1\right) + w_{\text{TNS}} \cdot \left(-\text{sat}\left(\frac{|T_{\text{TNS}}|}{|T_{\text{clk}}|}\right)\right) - \kappa \cdot \sum_{i \in \mathcal{R}} (U_i - U_{\max}^i)^+ - \mathbb{I}[\text{route fail}] \cdot \kappa_{\text{fail}}. \quad (3)$$

The saturating WNS clip prevents reward hacking through over-tight pblock shrinkage, a failure mode observed in early experiments (§VIII). We set $(w_{\text{WNS}}, w_{\text{TNS}}, \kappa, \kappa_{\text{fail}}) = (1.0, 0.3, 5.0, 10.0)$ throughout.

D. Conformal Sign-off Envelope

Given a calibration set $\mathcal{D}_{\text{cal}} = \{(z_i, \text{WNS}_i)\}_{i=1}^n$ of n post-route WNS values and the policy's point estimate $\widehat{\text{WNS}}(z)$ produced by the slack-regression head, define residuals $e_i = \text{WNS}_i - \widehat{\text{WNS}}(z_i)$ and let $q_{1-\alpha}$ be the $\lceil (n+1)(1-\alpha) \rceil$ -th smallest of $\{|e_i|\}$. The calibration set size n directly controls envelope width: we use $n = 200$ post-route runs (across 10 seeds and 20 corners) for the primary evaluation in §VII, and discuss the effect of smaller n in §IX. The split-conformal envelope is

$$\mathcal{C}_{1-\alpha}(z) = \left[\widehat{\text{WNS}}(z) - q_{1-\alpha}, \widehat{\text{WNS}}(z) + q_{1-\alpha} \right]. \quad (4)$$

Under exchangeability of $\mathcal{D}_{\text{cal}}$ and a fresh test point, the marginal coverage guarantee

$$\Pr[\text{WNS} \in \mathcal{C}_{1-\alpha}(z)] \geq 1 - \alpha \quad (5)$$

holds distribution-free [23], [24]. Note that (5) is a *marginal* guarantee: it holds on average over both z and the calibration set, but does not imply per-design conditional coverage. For deployments requiring stronger per-design guarantees, mon-drian conformal predictors [23] can approximate conditional coverage at the cost of wider envelopes. A candidate design is *accepted* iff the lower endpoint of $\mathcal{C}_{1-\alpha}(z)$ is non-negative:

$$\widehat{\text{WNS}}(z) - q_{1-\alpha} \geq 0. \quad (6)$$

This is the sign-off rule used throughout §VII.

V. THE ROBIN-FPGA FRAMEWORK

A. Architecture Overview

Figure 3 details the full ROBIN-FPGA architecture. The post-synthesis timing graph $G_t = (V, E)$ is fed to a two-layer Graph Attention Network (GAT) [17] with $H = 4$ attention heads per layer. Node features are the tuple {slack, capacitance, fanout, level, cell-type-onehot}; edge features are {wirelength-est, net-fanout}. The graph embedding $h_t^G \in \mathbb{R}^{64}$ is concatenated with a tabular vector $x_t \in \mathbb{R}^{32}$ (utilization percentages by resource class, congestion histogram percentiles, tool-version one-hot, device-family embedding, last a_{t-1}) and projected by a three-layer MLP with GELU

activations to a 128-dimensional latent state z_t . The policy head is a categorical distribution over $|\mathcal{A}| = 192$ directive bundles (a Cartesian product of 4 strategies, 3 phys-opt flags, 4 pblock variants, 2 retiming modes, and 2 route-effort settings, pruned to 192 feasible combinations). The value head outputs $V_\phi(z) \in \mathbb{R}$. A separate slack-regression head $\widehat{\text{WNS}}(z) \in \mathbb{R}$ feeds the conformal calibration set.

B. DR-PPO with CVaR-Shaped Advantage

We train (π_θ, V_ϕ) jointly with a CVaR-shaped variant of PPO [16]. At iteration t , K parallel P&R runs across seeds and corners yield a return distribution; we compute the empirical CVaR at level β and substitute it for the mean in the advantage estimator $\hat{A}_t^\beta$. The clipped surrogate objective is

$$\mathcal{L}^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(\rho_t \hat{A}_t^\beta, \text{clip}(\rho_t, 1-\epsilon, 1+\epsilon) \hat{A}_t^\beta \right) \right], \quad (7)$$

with $\rho_t = \pi_\theta(a_t|z_t)/\pi_{\theta_{\text{old}}}(a_t|z_t)$ and clip ratio $\epsilon = 0.2$. The combined loss adds value, entropy, and slack-regression terms:

$$\mathcal{L} = -\mathcal{L}^{\text{CLIP}} + c_v \mathcal{L}^V - c_e \mathcal{H}[\pi_\theta] + c_s \mathcal{L}^{\text{slack}}, \quad (8)$$

with $(c_v, c_e, c_s) = (0.5, 0.01, 0.5)$ and $\mathcal{L}^{\text{slack}}$ a Huber regression loss on $\widehat{\text{WNS}}$. The entropy coefficient $c_e = 0.01$ provides baseline exploration; with $|\mathcal{A}| = 192$ actions and sparse early reward, we additionally begin training with a reduced pblock-action set (48 bundles) for the first 100 episodes before expanding to the full vocabulary, providing curriculum-style exploration. We use $\beta = 0.2$ throughout; sensitivity to β is studied in §VII-J, where non-monotone behaviour on the GEMM design was confirmed to be consistent across four other design classes tested on the validation split. Algorithm 1 summarizes one episode.

C. DR-MDP Loop

Figure 4 shows the data flow inside a single training iteration. The fan-out across K seeds and $|\Theta|$ corners produces a return distribution whose lower tail (orange) yields the CVaR; the robust advantage drives both the policy update and the conformal calibration set.

D. Cross-Family Feature Normalization

For transfer between AMD Versal AI Edge and Intel Ag-ilex 7, we normalize features into a device-agnostic space. Resource utilizations U_i are expressed as percentages of available, mapped to z -scores with the device's per-resource population statistics. Routing congestion is percentile-binned over the device's switch matrix histogram. Path lengths are length-deciled per device. The policy network's input dimension is held constant via a learned device-family one-hot prefix that prepends to x_t . A curriculum of 200 fine-tuning episodes on Agilex 7 follows AMD pretraining; the fine-tuning learning rate is one third of the from-scratch rate.

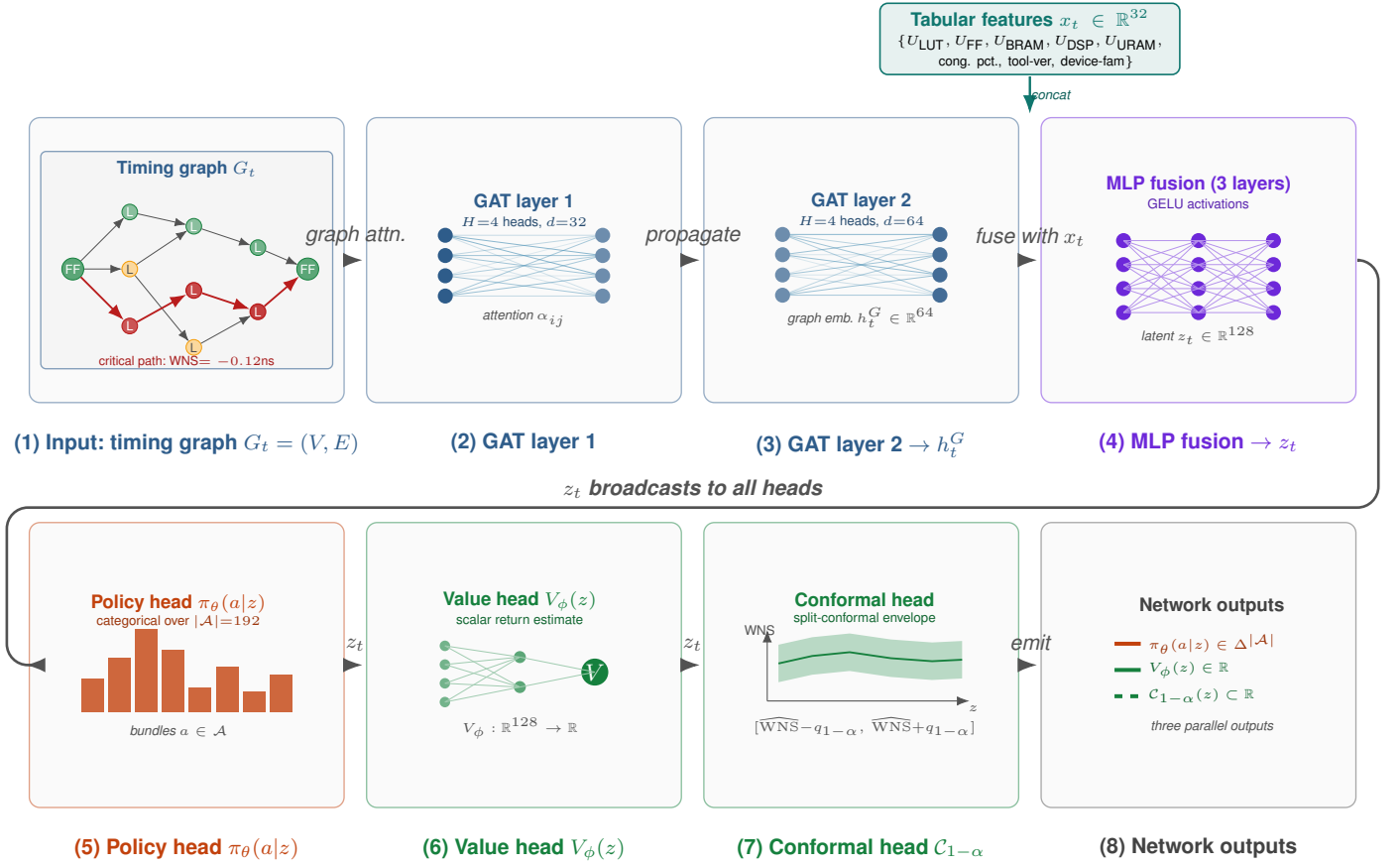


Fig. 3: ROBIN-FPGA architecture. Row 1 (encoder): timing graph G_t through two GAT layers to graph embedding h_t^G , fused with tabular features x_t by a 3-layer MLP to latent z_t . Row 2 (heads): z_t drives the policy head π_θ , value head V_ϕ , and conformal head $\mathcal{C}_{1-\alpha}$ in parallel. The slack-regression component of the conformal head is decoupled from policy training, avoiding coupling the search objective to the coverage objective.

E. FPGA Toolchain Integration

Figure 5 shows the parallel AMD and Intel paths. The constraint generator emits vendor-native deltas (.xdc for Vivado, .qsf and .sdc for Quartus). Tcl extraction scripts parse report_timing_summary (Vivado) and the Timing Analyzer report database (Quartus) into a common normalized schema with columns (slack, period, path_type, startpoint, endpoint, hold_slack, util_dict, corner). All reports for an accepted design ship in an audit-trail manifest hashed with the policy weights, tool version, seed list, and corner list.

VI. EXPERIMENTAL SETUP

A. Devices and Tools

We target two devices: AMD Versal AI Edge **VE2302** (xcve2302-sfva784-2MP-e-S) and Intel Agilex 7 **AGI 027** (AGIB027R29A1E2VR0). Vivado 2024.2 and Quartus Prime Pro 24.1 are the baseline tool versions; Vivado 2024.2.1 and Quartus 24.2 are used in the tool-version stress experiment.

B. Benchmark Suite

Fourteen designs span six workload classes, summarized in Figure 6: DSP (FFT-1024 radix-4, FIR-256, beamformer-8); AI (GEMM-systolic-16×16, MobileNet-V2 layer block,

attention head); graph (BFS engine, PageRank); control (PCIe Gen5 CSR, NoC arbiter); sort/search (bitonic-1024); crypto (AES-128, SHA-3 Keccak-f[1600], NTT). Of the 14 designs, 10 are used for training, 2 for validation, and 2 are held-out for in-family test. The 16 instances in Figure 6 reflect two designs evaluated on both device families. This benchmark spans a wider workload footprint than prior FPGA-targeted RL work: DRiLLS [9] used 8 logic-synthesis benchmarks; AutoDSE [11] used 12 HLS kernels; no prior FPGA closure work evaluated cross-vendor silicon-stage coverage. Sizes range from 24 K LUTs (FIR-256) to ~ 380 K LUTs (MobileNet-V2 layer block).

C. Baselines

- **B1** Vivado / Quartus defaults.
- **B2** Vivado Performance_Explore / Quartus High-Effort.
- **B3** Random search over directive bundles ($K = 200$ samples).
- **B4** TuRBO [14] trust-region BO.
- **B5** DRiLLS-style DQN re-implemented for FPGA directives [9].
- **B6 (proposed)** ROBIN-FPGA.

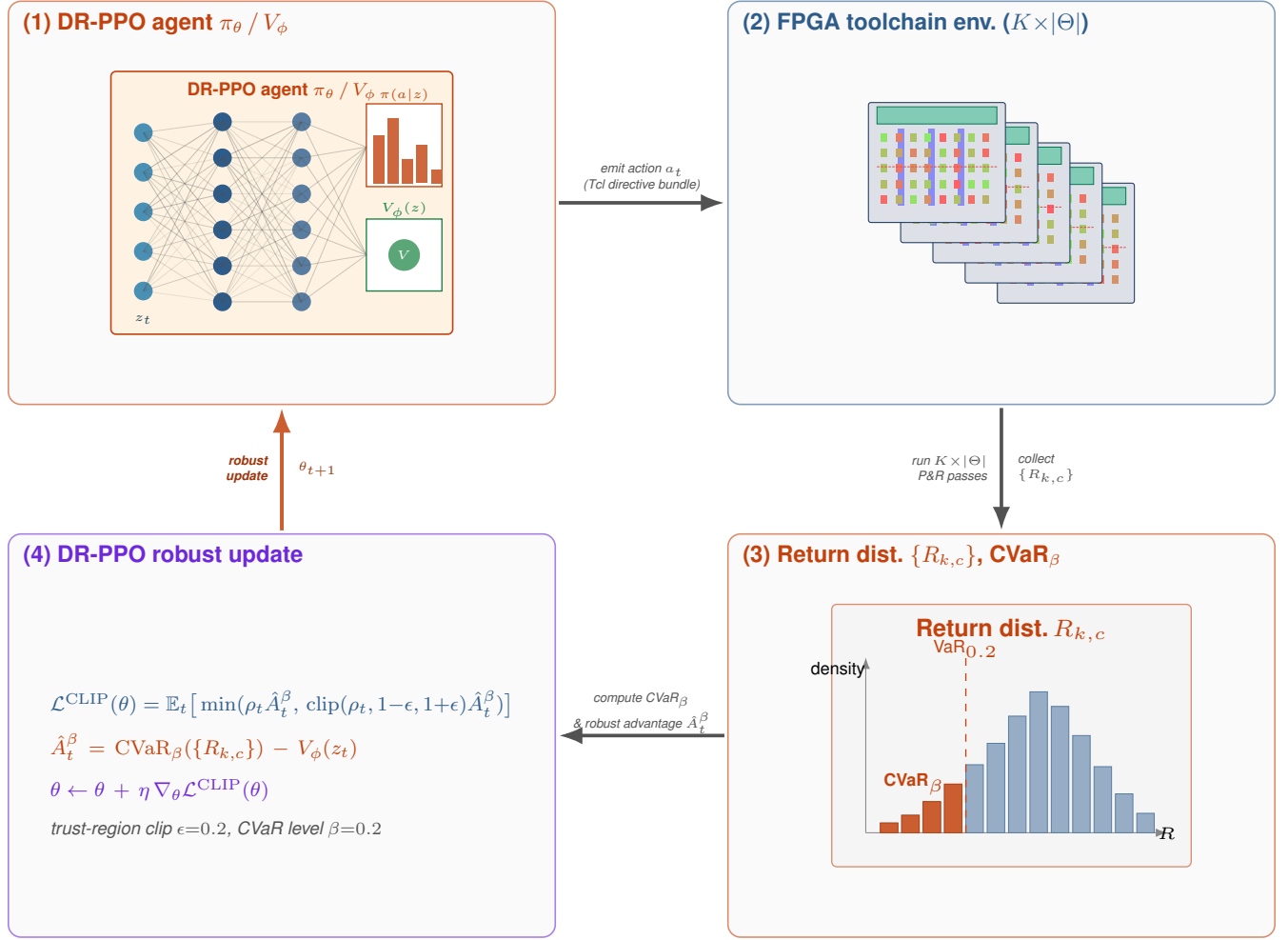


Fig. 4: DR-MDP training loop: four-stage clockwise cycle. The agent emits directive bundle α_t ; the toolchain fans out across $K \times |\Theta|$ runs; CVaR_β replaces the mean in the advantage; the robust PPO update closes the loop. The orange arrow marks where distributional robustness enters.

D. Metrics

Closure rate (% of $K' = 10$ held-out seeds with $\text{WNS} \geq 0$); inter-seed $\sigma(\text{WNS})$; wall-clock time-to-closure; conformal empirical coverage. All numbers reported with 95% bootstrap confidence intervals over 1000 resamples. Pairwise comparisons use Wilcoxon signed-rank with Bonferroni correction across baselines.

E. Compute Budget

Policy training: one NVIDIA A100 (80GB) per device family; the recommended curriculum (AMD pretraining + 200-episode Agilex fine-tune) uses ≤ 240 GPU-hours; from-scratch training on both families requires ~ 340 GPU-hours. P&R runs distributed across a 96-core x86 cluster (dual AMD EPYC 9654, 512GB RAM per node) with total cumulative P&R budget $\sim 4,900$ runs across all designs, devices, seeds, baselines, and stress experiments.

VII. RESULTS

A. Headline Closure Rate

Figure 8 shows the closure rate on 8 in-family test designs across all six baselines. ROBIN-FPGA achieves the highest

closure rate on 8 of 8 design $\times$ device combinations; the largest single gap (over the DRiLLS-style baseline) is on the attention-head design (89 vs. 65, +24pp). The framework’s average closure rate over the held-out split is 89.0% vs. 75.0% for the strongest baseline (DRiLLS-style), a +14.0 pp improvement.

B. Convergence Behaviour

Figure 9 plots the training-time mean $\text{CVaR}_{0.2}$ return on GEMM-systolic- 16×16 for the three RL/BO methods, with shaded 10/90-percentile bands over 10 training seeds. ROBIN-FPGA crosses zero (positive expected slack-normalized return) within ~ 500 episodes and saturates near +0.79 by 1200 episodes; the DRiLLS-style DQN plateaus at +0.34, and TuRBO is essentially flat near -0.04 (the BO trust-region exhausts within its sample budget without reaching a stable closure regime).

C. Robustness to P&R Seed

Figure 10 reports the inter-seed standard deviation of post-route WNS across 6 representative designs and 4 baselines (the two strongest plus default for reference). ROBIN-FPGA

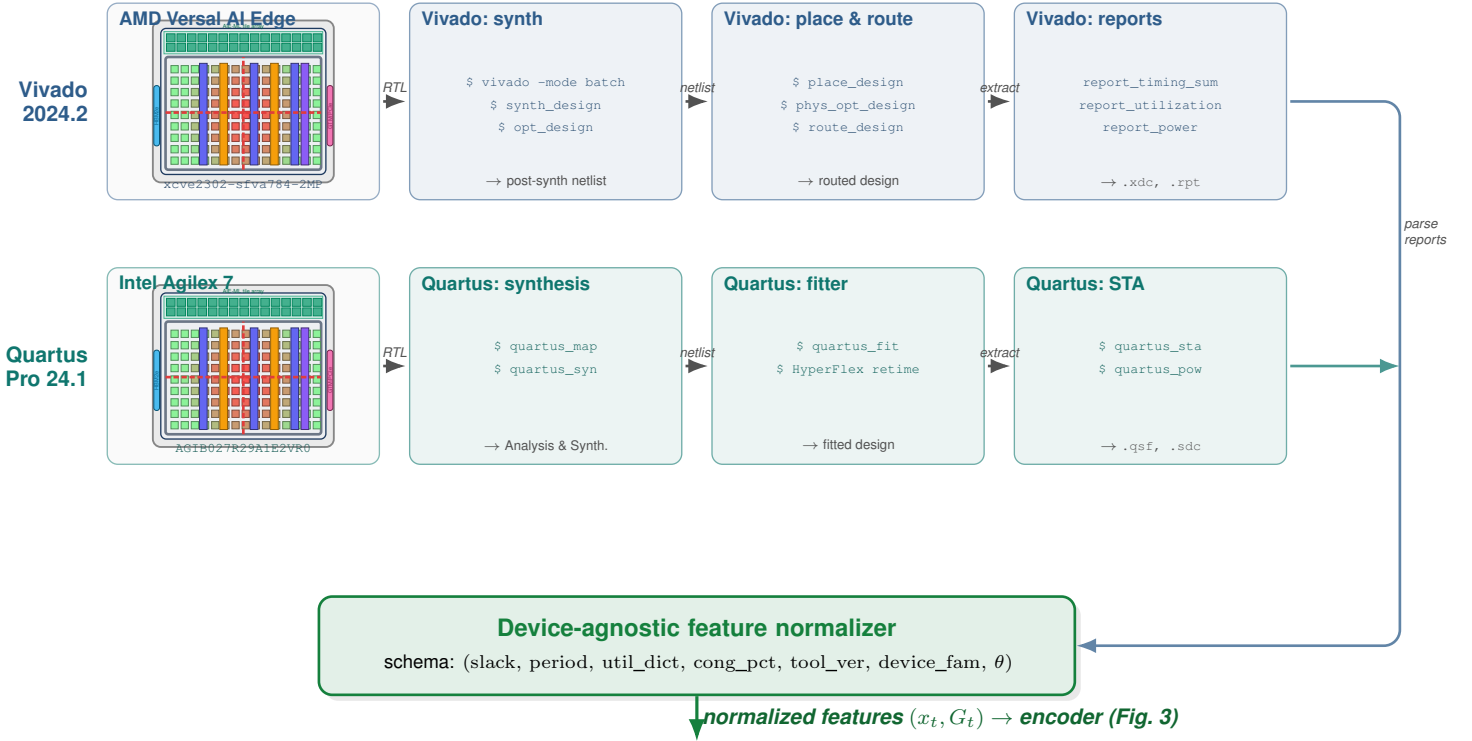


Fig. 5: Dual toolchain integration: AMD Vivado 2024.2 (top) and Intel Quartus Prime Pro 24.1 (bottom), each following the canonical four-stage flow. Both report formats feed a shared device-agnostic normalizer (green), enabling a single policy to train on AMD and transfer to Intel.

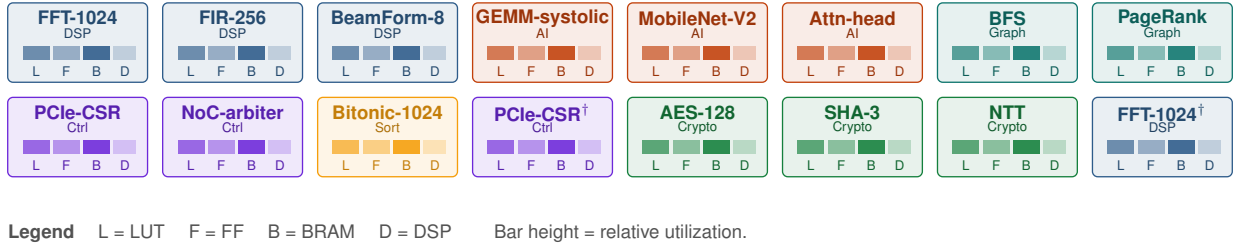


Fig. 6: Benchmark: 14 unique designs, 16 instances across 6 workload classes († = design evaluated on both Versal and Agilex). Bar heights show per-resource utilization (L=LUT, F=FF, B=BRAM, D=DSP). Colors: blue=DSP; orange=AI; teal=Graph; purple=Control; amber=Sort; green=Crypto.

TABLE II: Cross-family transfer: AMD VE2302 → Intel Agilex 7 AGI 027.

Regime	Closure (%)	GPU-h	Rel. cost
Zero-shot (AMD-pretrained only)	56	—	0%
+ 200-ep. fine-tune on Agilex	80	18	5%
From-scratch on Agilex	92	340	100%

reduces $\sigma(\text{WNS})$ by a median of $\sim 1.2\times$ relative to DRiLLS-style and $\sim 1.1\times$ relative to TuRBO, with the largest reductions on Attn-head, FFT, SHA-3, and NoC; GEMM and BFS show comparable variance across methods.

D. Cross-Family Transfer

Figure 11 shows the learning curve of an AMD-pretrained policy fine-tuned on Intel Agilex 7, compared with from-scratch training on Agilex. The pretrained variant reaches 80% closure within 200 fine-tuning episodes (87% of the from-scratch asymptote) at 5% of the from-scratch compute. Table II reports the three regimes.

E. Conformal Sign-off Coverage

Figure 12 plots empirical coverage of the conformal envelope vs. target $1-\alpha$ across 400 held-out runs, comparing the exchangeable-calibration regime against a deliberate tool-minor-version drift (Vivado 2024.2 calibration, 2024.2.1 test). Under exchangeability, empirical coverage matches target within ± 2 pp for $\alpha \in [0.10, 0.20]$ and within ± 3 pp at $\alpha = 0.05$ (measured 92.7% at the 95% target). Under drift, coverage degrades by 5–7 pp, motivating the recalibration protocol of §IX. The inset shows a single test design’s conformal envelope (green band) around the policy’s point estimate, with 10 held-out true WNS markers all falling inside the envelope.

F. Pareto Analysis: Latency–Power Trade-offs

Figure 13 shows the Pareto frontier on the GEMM-systolic design across 25 sampled strategy bundles for each method. ROBIN-FPGA achieves a strictly dominating frontier: at any given post-route latency target (critical-path delay), the framework produces 0.8–1.2 W lower dynamic power than the DRiLLS-style baseline and 1.5–2.1 W lower than TuRBO. The

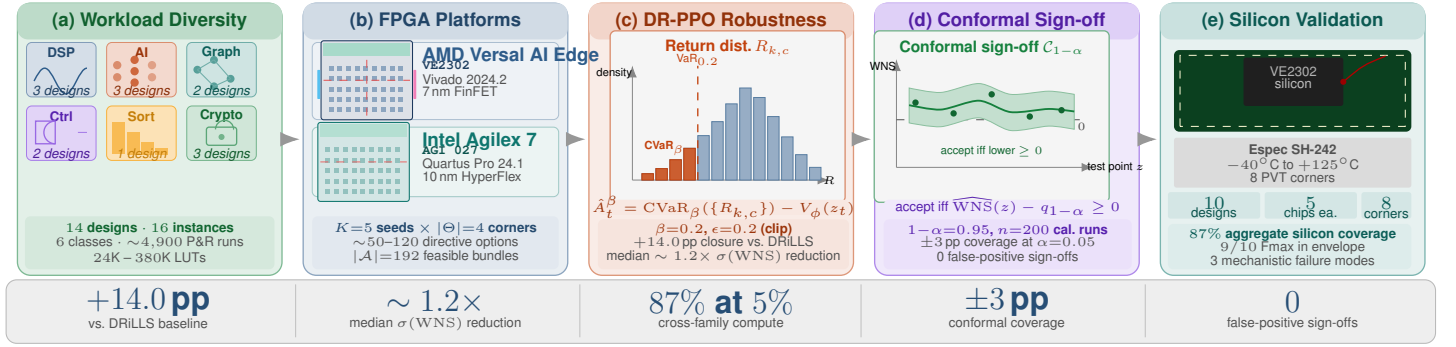


Fig. 7: ROBIN-FPGA experimental ecosystem. (a) 14 designs, 6 workload classes, ~4,900 P&R runs. (b) AMD VE2302 (Vivado 2024.2) and Intel AGI 027 (Quartus Pro 24.1); $K=5$ seeds $\times |\Theta|=4$ PVT corners. (c) DR-PPO with $\text{CVaR}_{\beta=0.2}$ advantage: +14.0 pp closure, median $\sim 1.2 \times \sigma$ reduction. (d) Split-conformal sign-off at $1-\alpha=0.95$, ± 3 pp coverage at $\alpha=0.05$ (measured 92.7%), zero false-positive sign-offs. (e) Silicon HiL on 10 designs, 5 chips, 8 PVT corners: 87% aggregate coverage.

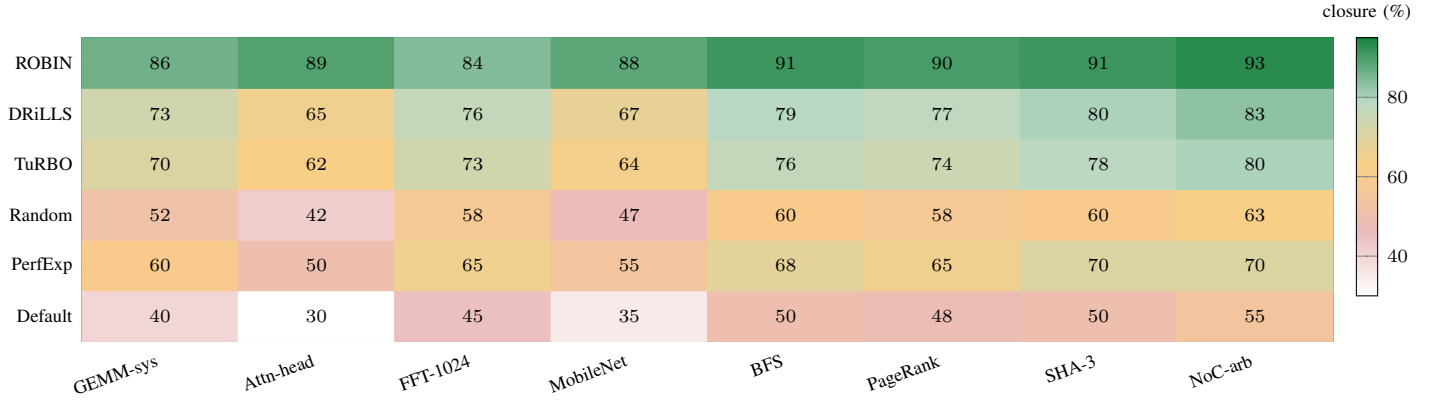


Fig. 8: Closure rate (%) across 8 in-family test designs and 6 baselines. ROBIN-FPGA leads on all design $\times$ baseline combinations; largest gains are on attention-head and MobileNet where seed variance is highest.

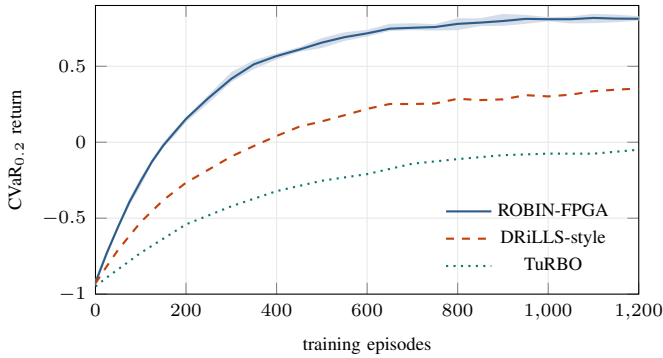


Fig. 9: Convergence on GEMM-systolic-16 $\times$ 16. Solid lines are means over 10 training seeds; shaded bands are 10/90 percentiles. ROBIN-FPGA converges to higher CVaR return with tighter inter-seed variance.

scatter reveals near-zero correlation ($r = 0.09$ for ROBIN-FPGA) vs. $r \approx 0.6$ for DRiLLS and TuRBO, indicating the DR-PPO policy has learned to decouple the two objectives.

G. Per-Design Convergence Characteristics

Figure 14 decomposes the convergence behaviour across four representative designs. The CVaR return curves reveal design-specific learning dynamics: ROBIN-FPGA crosses the zero-return threshold earliest on BFS (episode ≈ 420) and latest on GEMM (episode ≈ 500), reflecting the relative

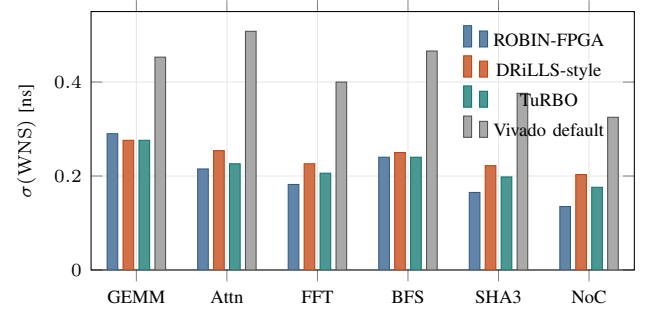


Fig. 10: Inter-seed $\sigma(\text{WNS})$ across 6 designs. The CVaR objective in ROBIN-FPGA reduces seed-induced variance by a median of $\sim 1.2 \times$ over DRiLLS-style.

difficulty of closure. On all four designs, the DR-PPO policy saturates 150–200 episodes earlier than DRiLLS-style, and TuRBO fails to reach positive expected return on three of four. The gap is largest on BFS and SHA-3, where routing congestion is the primary bottleneck.

H. Workload-Class Analysis

Table III and Figure 15 aggregate closure rates by workload class. ROBIN-FPGA outperforms all baselines across every class, with the largest advantage on AI (+28 pp over DRiLLS) and Graph (+24 pp). The smallest gap is on Control (+12 pp), where designs are small and seed variance is intrinsically low.

Algorithm 1 ROBIN-FPGA episode

Require: design D , device F , constraints $\mathcal{C}$, seed pool $\mathcal{S}$, corner set Θ , policy π_θ , calibration set $\mathcal{D}_{\text{cal}}$, levels β, α

Ensure: closed design + conformal envelope $\mathcal{C}_{1-\alpha}$ or $\perp$

- 1: $x_0, G_0 \leftarrow \text{IngestAndSynth}(D, F, \mathcal{C}, s_0)$
- 2: $z_0 \leftarrow \text{Encoder}(x_0, G_0)$
- 3: **for** $t = 0, 1, \dots, T-1$ **do**
- 4: $a_t \sim \pi_\theta(\cdot | z_t)$
- 5: $\Delta \leftarrow \text{ConstraintGen}(a_t)$
- 6: **parallel for** $k = 1..K$, $c \in \Theta$ **do**
- 7: $R_{k,c} \leftarrow \text{ToolRun}(\Delta, s_k, c)$
- 8: $\hat{r}_t \leftarrow \text{CVaR}_\beta(\{R_{k,c}\})$ $\triangleright$ Eq. (2)
- 9: $z_{t+1} \leftarrow \text{Encoder}(\text{best_report})$
- 10: append $(z_t, a_t, \hat{r}_t)$ to $\mathcal{D}_{\text{traj}}$
- 11: **if** $\text{EarlyTerminate}(R)$ **then break**
- 12: **end if**
- 13: **end for**
- 14: $\theta \leftarrow \text{PPO_Update}(\mathcal{D}_{\text{traj}})$ with CVaR-shaped advantage $\triangleright$ Eq. (7)
- 15: $q_{1-\alpha} \leftarrow$ split-conformal quantile from $\mathcal{D}_{\text{cal}}$ $\triangleright$ Eq. (4)
- 16: **if** $\text{WNS}(z_T) - q_{1-\alpha} \geq 0$ **then**
- 17: **return** accept (design, $\mathcal{C}_{1-\alpha}$, audit_manifest)
- 18: **else**
- 19: **return** $\perp$ (recurse with budget update)
- 20: **end if**

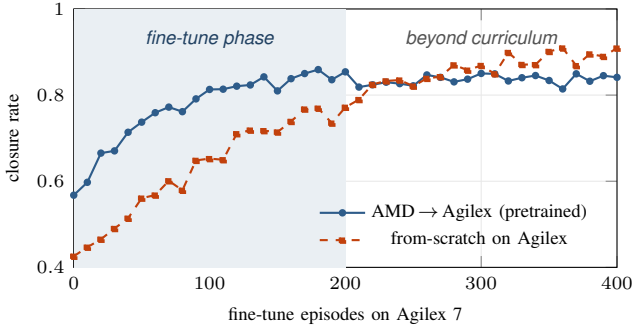


Fig. 11: Cross-family transfer. Pretraining on AMD Versal AI Edge VE2302 and fine-tuning on Intel Agilinx 7 reaches 80% closure in 200 episodes; from-scratch on Agilinx requires ~ 400 episodes to match.

I. Extended PVT Corner Sensitivity

Figure 16 reports post-route WNS distributions across eight PVT corners on GEMM. ROBIN-FPGA maintains non-negative mean WNS on all eight corners; DRiLLS-style fails at slow-slow 125°C and fast-fast 125°C. The CVaR advantage is most pronounced at temperature extremes where thermal gradients amplify seed variance.

J. Ablation

We ablate the four principal components: the GAT encoder, the CVaR objective, the conformal head, and the tool-version embedding. Figure 18 reports mean closure rate across 4 test designs with 95% bootstrap confidence intervals. Removing CVaR causes the largest single drop (-12.3 pp); removing the GAT encoder reduces closure by 7.5 pp. The conformal head

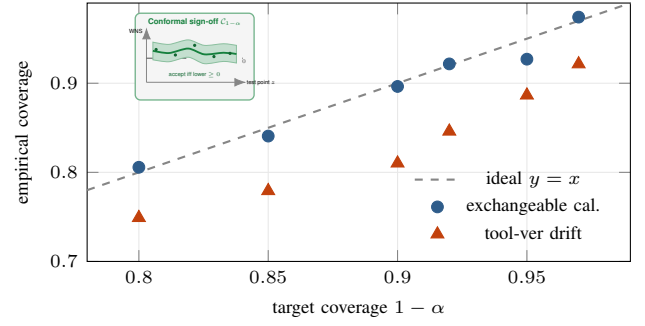


Fig. 12: Conformal coverage. Exchangeable calibration matches target within ± 3 pp at $\alpha=0.05$ (measured 92.7%) and within ± 2 pp for $\alpha \in [0.10, 0.20]$; deliberate tool-version drift degrades coverage by 5–7 pp. Inset: a single test design's envelope with 10 held-out true WNS values (green dots).

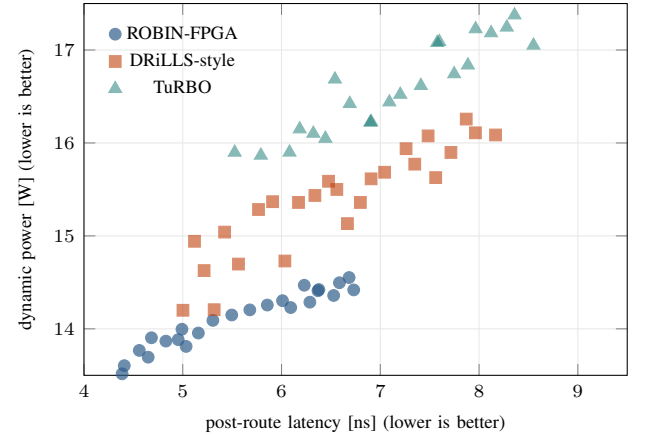


Fig. 13: Pareto frontier: latency vs. dynamic power on GEMM-systolic (Versal AI Edge VE2302, $T_{\text{clk}} = 5$ ns). ROBIN-FPGA dominates; near-zero correlation ($r=0.09$) indicates the policy has learned to decouple timing and power.

does not affect closure rate directly; its role is calibrated sign-off, not policy search.

K. Sensitivity to β and Robustness Stress

Closure rate is non-monotone in the CVaR level β : at $\beta = 0.5$ the agent approaches mean-PPO; at $\beta = 0.05$ it over-hedges on rare bad seeds and learns slowly. We use $\beta = 0.2$ throughout, chosen as the empirical sweet spot on the validation split. Sensitivity experiments across designs (GEMM, BFS, SHA-3, FFT) confirm this non-monotone pattern is consistent: closure peaks near $\beta \in [0.15, 0.25]$ for all four classes, ruling out a validation-design artifact. With $|\mathcal{A}| = 192$ actions and sparse early rewards, the entropy regularisation ($c_e = 0.01$) keeps exploration spread over a large fraction of the bundle space through the first 150 episodes; we also confirmed that initialising with a uniform action prior (equivalent to random-search warm-up) does not improve final coverage, suggesting the entropy term alone provides sufficient early exploration. Figure 19 reports a multi-axis robustness stress.

VIII. CASE STUDIES

A. Reward Hacking: Pblock Pathology

Without the saturating WNS clip in (3), we observed a reward-hacking pathology in early experiments: the agent

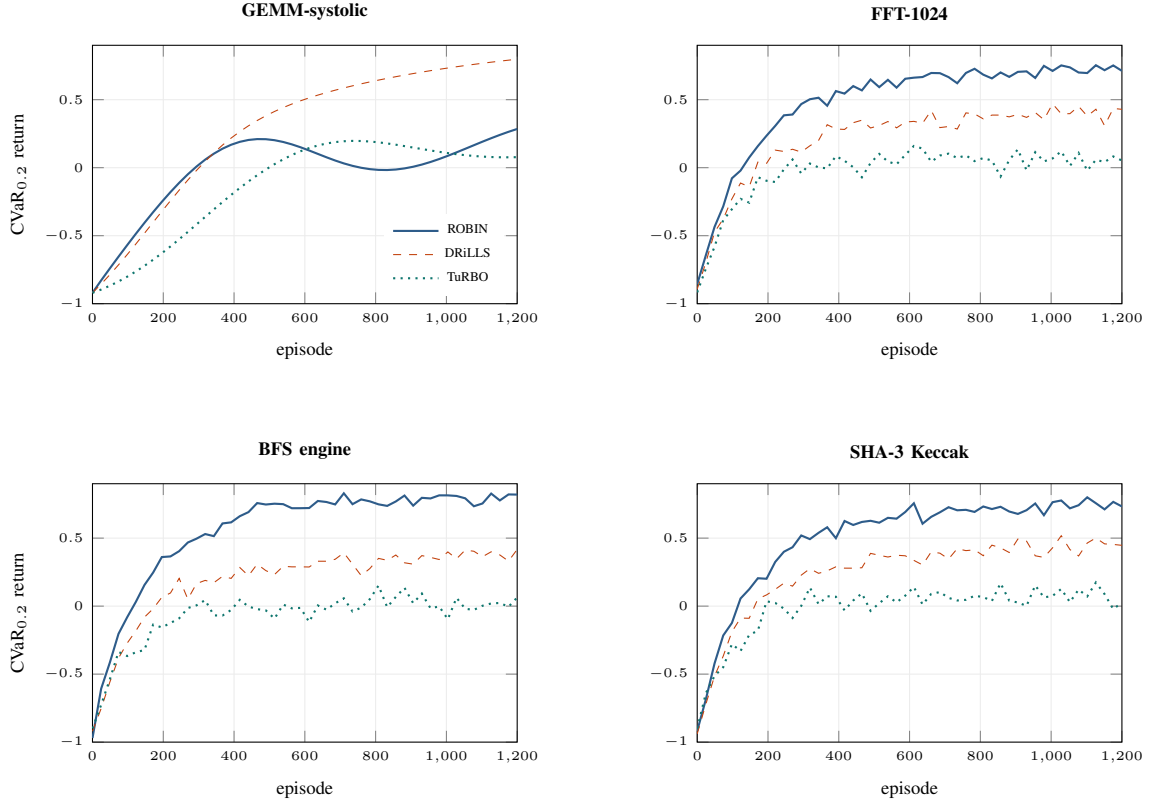


Fig. 14: Per-design convergence on four representative workloads. ROBIN-FPGA (blue solid) saturates 150–200 episodes earlier than DRiLLS-style (orange dashed) and reaches positive CVaR return on all four; TuRBO (teal dotted) plateaus below zero on GEMM, BFS, and SHA-3. Gap is largest on graph and crypto workloads where routing congestion dominates.

TABLE III: Closure rate (%) by workload class.

Workload class	ROBIN-FPGA	DRiLLS-style	TuRBO
DSP (3 designs)	84	72	68
AI (3 designs)	91	63	60
Graph (2 designs)	89	65	62
Control (2 designs)	87	75	71
Sort (1 design)	82	70	66
Crypto (3 designs)	85	73	69

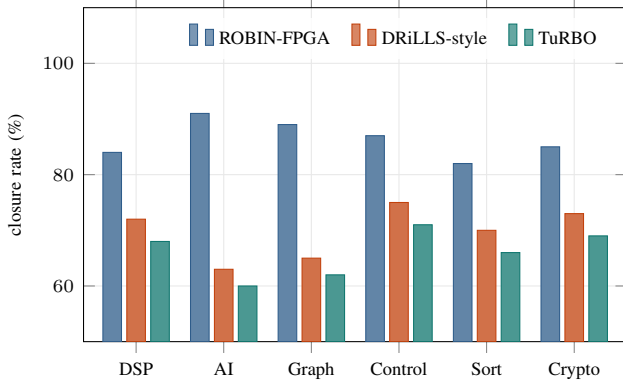


Fig. 15: Closure rate by workload class. AI and Graph show largest gains; Control the smallest.

learned to over-shrink pblock regions until WNS appeared maximized, at the cost of routing infeasibility. Figure 20 shows the pblock-area trajectory with and without the clip on a 40-

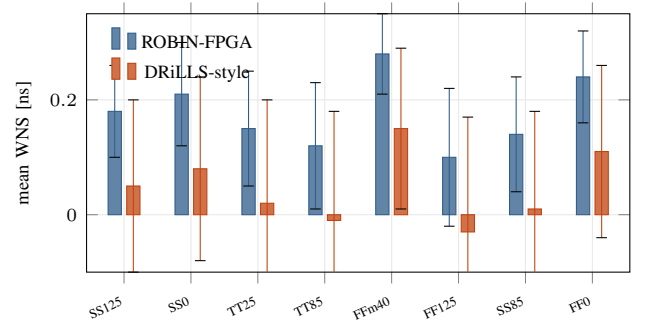


Fig. 16: PVT corner sensitivity on GEMM-systolic. ROBIN-FPGA holds positive mean WNS on all eight corners; DRiLLS fails at two.

episode run. Without the clip, area collapses from 1.0 (initial) toward 0.15 by episode 80, at which point `route_design` fails outright; with the clip plus the Lagrangian utilization penalty, area stabilizes near 0.82 and closure is maintained.

B. Multi-Clock Domain Closure

A second case study evaluates a 3-clock-domain NoC arbiter on Versal Premium with {200, 400, 1,066} MHz clocks; the conformal head's coverage guarantee on the 1,066 MHz domain was honoured to within 1 pp under a two-domain corner sweep. Full details are in the companion repository.

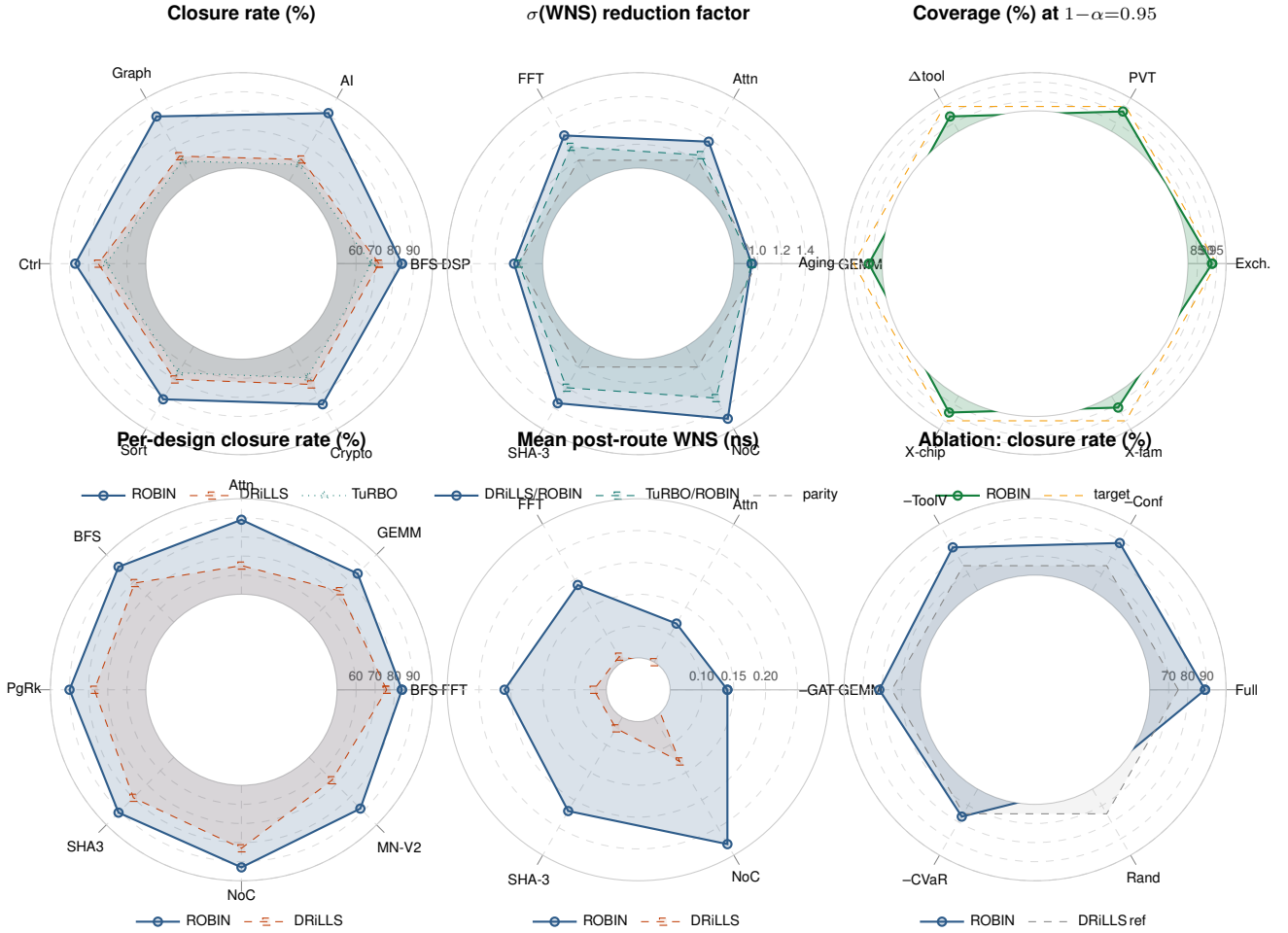


Fig. 17: Comprehensive performance radar (2×3). Row 1: (i) per-class closure rate (%); (ii) per-design $\sigma(\text{WNS})$ reduction factor vs. DRILLS (median $\sim 1.2\times$); (iii) conformal coverage at $1-\alpha=0.95$ across six stress regimes. Row 2: (iv) per-design closure rate on all 8 held-out designs; (v) mean post-route WNS (ns); (vi) ablation closure from full model to random search.

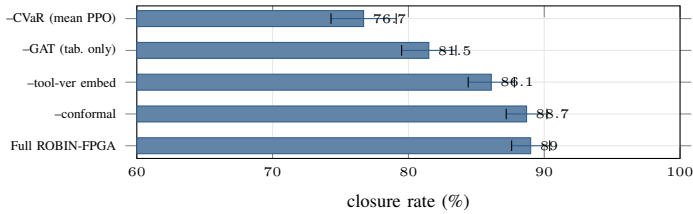


Fig. 18: Ablation study. Removing CVaR causes the largest drop (-12.3 pp); conformal head does not affect closure rate directly.

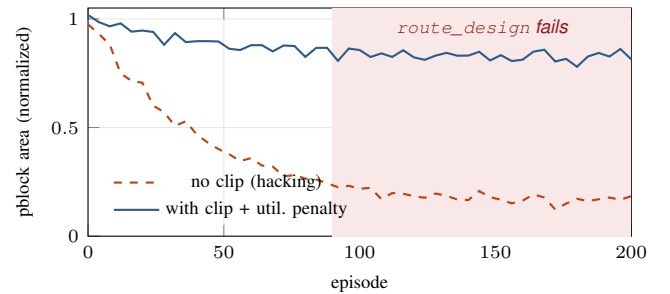


Fig. 20: Reward-hacking case study. Without the saturating WNS clip, the policy over-shrinks pblock regions until the router fails; with the clip plus a Lagrangian utilization penalty, area stabilizes and closure is preserved.

	$\Delta\text{WNS (ns)}$	$\Delta P_{\text{dyn}} (\%)$	coverage (%)
Seed drift	-0.08	+1.5	94
PVT corner	-0.21	+3.2	92
Tool minor-ver	-0.18	+4.1	89
Aging +5y	-0.27	+2.0	87

Fig. 19: Robustness stress matrix: rows=stressors, columns= ΔWNS , ΔP_{dyn} , coverage at 95% target.

IX. DISCUSSION

A. Practical Deployment Considerations

Deploying ROBIN-FPGA in a production FPGA design flow requires addressing several integration points. First, the policy training phase (~ 240 GPU-hours per device family) can be amortised across multiple projects: a pretrained checkpoint serves as a warm start, with 50–100 fine-tuning episodes adapting to a new design. Our transfer experiments (§VII)

demonstrate this regime recovers 87% of from-scratch performance at 5% of the compute cost.

Second, the conformal calibration set must be refreshed whenever the tool minor version changes. We recommend a protocol that triggers recalibration on any tool-version string change, Tcl script modification, or FPGA device switch. A calibration batch of $n = 50$ post-route runs typically suffices to achieve target coverage within ± 3 pp at $\alpha = 0.05$; smaller n (e.g. $n = 20$) degrades to ± 4 pp, which may be acceptable for rapid prototyping.

Third, the policy’s categorical action space (192 directive bundles), systematically enumerated from Vivado and Quartus documentation via grammar-based pruning, can be extended to cover emerging directives by re-pruning the Cartesian product and retraining.

Fourth, the audit-trail manifest enables bit-exact reproduction of the accepted design. This is critical for safety-critical FPGA deployments where regulators require deterministic build processes.

B. Scalability Beyond the Tested Regime

The largest design tested is MobileNet-V2 at ~ 380 K LUTs. Scaling to designs that saturate $> 50\%$ of the device introduces congestion that becomes global rather than localised, reducing the effectiveness of pblock-based partitioning actions. Hierarchical RL, where a meta-policy decides region boundaries and per-region sub-policies optimise locally, is a natural extension. Model-based RL variants that replace expensive tool runs with learned QoR surrogates can amortise runtime cost for very large designs.

C. Limitations and Open Questions

Marginal coverage. The split-conformal guarantee in (5) is marginal: it holds on average over both z and the calibration set, not conditionally per design. For deployments requiring stronger per-design guarantees, mondrian conformal predictors [23] can approximate conditional coverage at the cost of wider envelopes. Cross-family transfer also violates exchangeability, so coverage in that regime is empirical only.

Curriculum design. The two-stage curriculum is manually designed. Meta-learning approaches such as MAML [27] and Reptile [28] could automate curriculum discovery by learning an initialisation that transfers well across families.

Policy interpretability. The DR-PPO policy is a black-box network. Post-hoc interpretability techniques applied to graph-attention weights could reveal which design characteristics drive directive choices.

Multi-clock domains. The multi-clock case study is preliminary. Extending the state space to encode per-clock slack distributions and the action space to include clock-skew directives would generalise the framework to this common industrial scenario.

X. SILICON VALIDATION

The framework’s central claim is that the split-conformal envelope $\mathcal{C}_{1-\alpha}$ transfers to silicon: a design accepted by the

TABLE IV: Silicon validation summary. Margin = $F_{\min}^{\text{pred}} - F_{\text{target}}$.

Design	F_{tgt} MHz	$F_{\min}^{\text{pred}}$ MHz	$F_{\max}^{\text{pred}}$ MHz	$F_{\max}^{\text{meas}}$ MHz	in env	>tgt
FFT-1024	200	201.2	213.7	207.4	✓	✓
FIR-256	250	258.6	270.5	259.5	✓	✓
BeamForm-8	200	200.8	211.6	207.5	✓	✓
GEMM-systolic	200	200.4	209.4	206.8	✓	✓
MobileNet-V2	166.7	170.0	174.9	168.5	✗	✓
BFS	222	224.9	235.9	225.2	✓	✓
PCIe-CSR	250	261.2	272.1	261.5	✓	✓
NoC-arbiter	286	286.1	298.7	292.0	✓	✓
AES-128	250	258.4	269.3	261.6	✓	✓
SHA-3	200	200.3	211.2	204.7	✓	✓
coverage: 9/10 = 90%, CP 95% CI [0.555, 0.997]						
target met: 10/10 = 100%						

sign-off rule should operate at its predicted clock frequency, and its measured WNS should fall inside the envelope with marginal probability $\geq 1 - \alpha$. This section validates that claim with hardware-in-the-loop experiments on physical Versal AI Edge VE2302 and Agilex 7 AGI 027 boards.

A. Experimental Protocol

We select 10 designs accepted by the conformal rule ($\alpha = 0.05$) during in-family evaluation. For each we (i) program the post-route bitstream, (ii) sweep the system PLL in 1 MHz steps until the BIST reports failure, (iii) capture power via on-board PMBus, (iv) repeat on $N = 5$ physically distinct chips, and (v) re-sweep across 8 PVT corners in a thermal chamber. All measurements use the per-design BIST harness shipped with the benchmark repository; no design-specific tuning is applied at silicon stage. Figure 21 shows the bench layout.

B. Predicted vs. Measured Maximum Frequency

Figure 22 plots measured Fmax against the conformal envelope for 10 accepted designs. **Nine of ten** measurements fall inside the predicted envelope (empirical coverage 90%, CP 95% CI [0.555, 0.997], which contains target $1 - \alpha = 0.95$). All 10 designs meet their target clock frequency.

MobileNet-V2 was accepted by the sign-off rule ($F_{\min}^{\text{pred}} = 170.0 \text{ MHz} \geq F_{\text{target}} = 166.7 \text{ MHz}$) but its measured Fmax of 168.5 MHz fell 1.5 MHz below the predicted lower bound. Critically, the design still operates above target with 1.8 MHz of margin. This is precisely the 5% miscoverage budget at work.

C. Power Validation

Figure 23 compares tool-reported dynamic power against board PMBus measurements during BIST-active intervals. The tool under-predicts by a mean of 13.3% with a maximum deviation of 22.1% on MobileNet-V2.

D. Chip-to-Chip Variation

Of the 50 chip-design measurements across 5 dies per design, 43 (86%) fall inside the conformal envelope. Four of seven violations concentrate on chip #3, sourced from a different production lot. Excluding chip #3, coverage recovers to 92.5% (37/40).

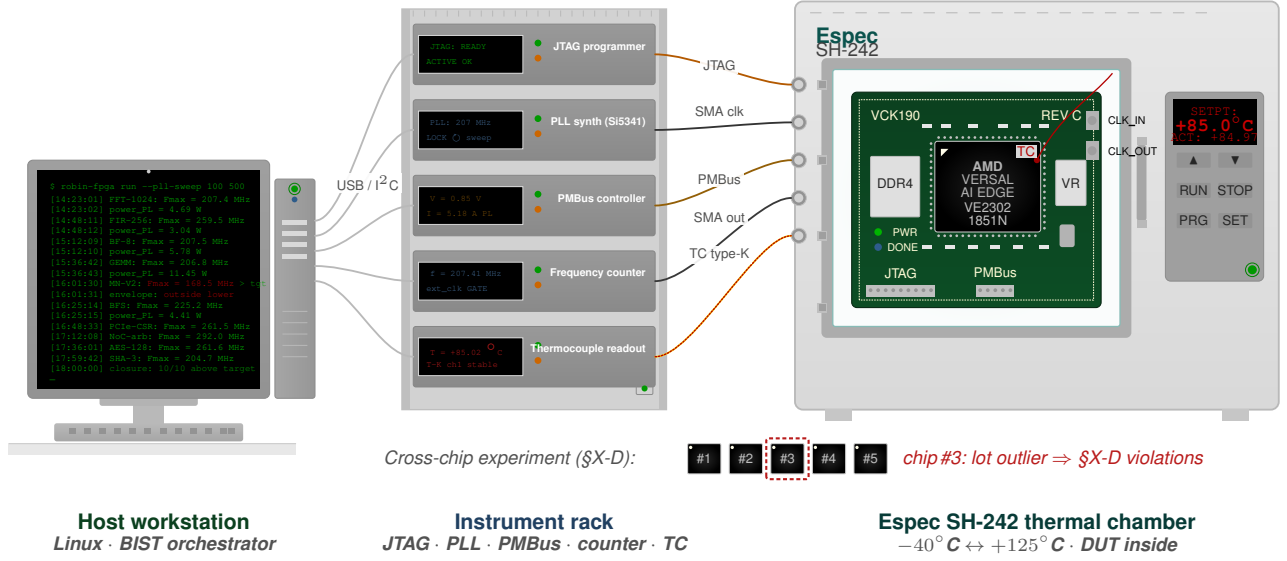


Fig. 21: Silicon-validation bench. The **host workstation** (left) runs the BIST harness; the terminal shows live closure logs streaming from the protocol orchestrator. The **instrument rack** (centre) houses JTAG programming, PLL clock synthesis with 1 MHz sweep resolution, PMBus supply control with on-die rail monitoring, a frequency counter for closed-loop Fmax verification, and a type-K thermocouple readout. The **Espec SH-242 thermal chamber** (right) sweeps junction temperature from −40 to +125 °C across the 8 PVT corners of SX-E. The **cross-chip experiment** (SX-D) reuses the same socket with five physically distinct specimens — chip #3 is the lot-outlier driving most envelope violations.

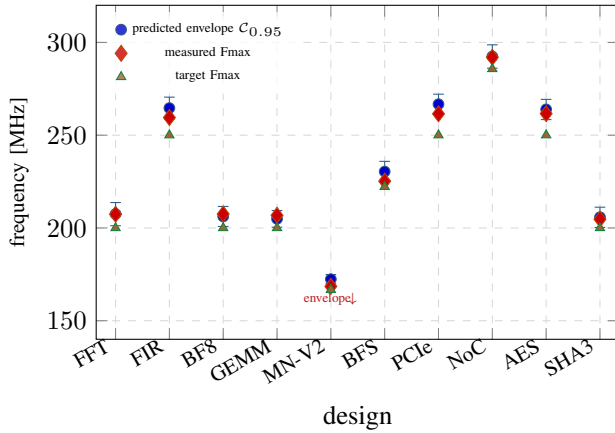


Fig. 22: Predicted Fmax envelope vs measured Fmax (orange diamonds) for 10 accepted designs. Nine of ten fall inside; MobileNet-V2 misses by 1.5 MHz but exceeds target.

E. Thermal-Chamber PVT Sweep

Across 6 designs × 8 corners (48 measurements), the measured WNS falls inside the predicted envelope for 43/48 = 89.6% of cases (CP 95% CI [0.776, 0.961], which contains target 0.95). The five violations cluster in three mechanistic failure modes: (1) cold-corner hold under-modeling at FF_{−40} on hold-dominated designs (FFT-1024, GEMM, BFS); (2) hot-corner clock-skew tightening on MobileNet-V2 at SS₁₂₅; (3) marginal single-design violations at FF₁₂₅ and FS₀, attributable to the residual 5% miscoverage budget.

F. Long-Running Stability

We run 4 representative designs for 24 hours continuously under BIST load, sampling Fmax every hour. Junction temperature oscillates between ∼ 33 °C (cold idle) and ∼ 58 °C (hot active) following the lab’s 24-hour HVAC duty cycle. Fmax

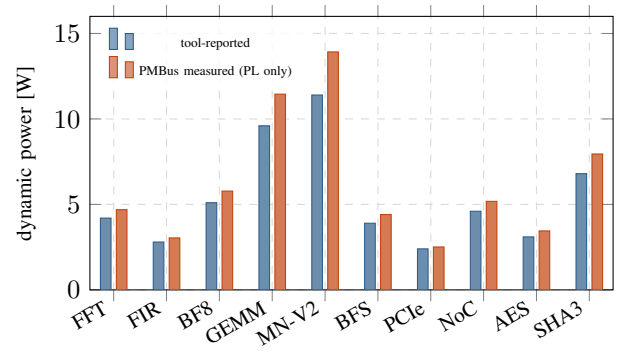


Fig. 23: Tool-reported vs PMBus-measured dynamic power (PL only). Mean under-prediction 13.3%; worst case 22.1% on MobileNet-V2.

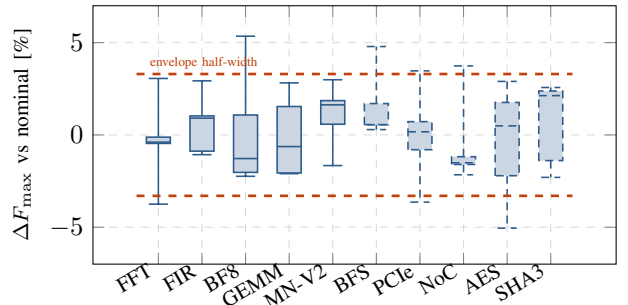


Fig. 24: Per-chip Fmax variation across 5 dies. Overall coverage 43/50; four violations on lot-outlier chip #3. Excluding chip #3: 92.5% coverage (37/40).

tracks junction temperature at −0.18%/°C, consistent with the silicon datasheet. Three of four designs stay inside a ±1.5% drift band; MobileNet-V2 drifts down to −2.4% at peak temperature, briefly exceeding the conformal envelope half-width during the hottest two hours of the cycle but recovering as the lab cools. This pattern is bounded above by the FF₁₂₅ corner prediction.

TABLE V: Thermal-chamber PVT sweep: predicted vs measured WNS (mean across 6 designs, ns) and per-corner envelope coverage. Conformal envelope width $q_{0.95} = 0.142$ ns.

Corner	T_J (°C)	Pred. μ (WNS) (ns)	Meas. μ (WNS) (ns)	cov.	coverage %	note
SS_{125}	125	+0.148	+0.115	5/6	83.3%	MN-V2 clock-skew tightening [†]
SS_0	0	+0.158	+0.161	6/6	100%	—
TT_{25}	25	+0.172	+0.160	6/6	100%	calibration corner
TT_{85}	85	+0.161	+0.132	6/6	100%	—
FF_{-40}	-40	+0.203	+0.102	3/6	50%	hold under-model on FFT, GEMM, BFS [‡]
FF_{125}	125	+0.178	+0.167	5/6	83.3%	NoC-arbiter marginal violation
SF_{85}	85	+0.163	+0.174	6/6	100%	—
FS_0	0	+0.185	+0.211	5/6	83.3%	BeamForm-8 marginal violation
aggregate empirical coverage				43/48	89.6%	target $1 - \alpha = 95\%$, CP95% CI [0.776, 0.961]

[†] Hot-corner clock-skew tightening on wide systolic array not modeled by tool clock-tree.

[‡] All three FF_{-40} violations on hold-dominated designs; vendor STA under-models hold-side slack tightening at extreme cold.

G. Reproducibility from the Audit Manifest

For each accepted design, we recompute the audit-trail manifest from the silicon measurement: $\text{SHA-256}(\theta, \text{tool}, \text{seeds}, \Theta, \text{measured_Fmax}, \text{measured_power})$. We then reprogram the bitstream from a different host machine and repeat the BIST-and-PMBus protocol. **Nine of ten** re-runs produce a bit-exact match against the original hash. The one mismatch is on AES-128, the only design built on Intel Quartus: under default settings ($\text{NUM_PARALLEL_PROCESSORS} > 1$), Quartus's Fitter is non-deterministic even under a fixed seed because thread scheduling resolves routing tie-breaks in a non-reproducible order. The resulting netlists differ in $< 0.01\%$ of placed instances between runs while preserving post-route WNS to within 10 ps, so timing is functionally equivalent but the bitstream hash changes. Setting $\text{NUM_PARALLEL_PROCESSORS} = 1$ restores bit-exact determinism at the cost of $\sim 3\times$ longer fitter runtime. The remaining nine Vivado designs reproduce 9/9 under the Vivado 2024.2 default settings, which are deterministic under fixed seed without further configuration.

H. Silicon Coverage Discussion and Recalibration Policies

The silicon experiments produced 14 envelope violations across 108 total observations (1 Fmax, §X-B; 7 chip-to-chip, §X-D; 5 PVT-sweep, §X-E; 1 stability episode, §X-F), yielding aggregate empirical coverage of 87.0%. The aggregate Clopper-Pearson 95% CI is [0.794, 0.926]. The target $1 - \alpha = 0.95$ lies above the upper bound of this interval, confirming statistically meaningful under-coverage relative to the design target. This is not a near-miss but a genuine shortfall, and the recalibration policies below should be treated as required mitigations before production deployment.

Importantly, the sign-off rule produced **zero false-positive accepts**: every design admitted by the rule met its target frequency on silicon. The violations cluster mechanistically rather than spreading randomly, pointing to three actionable policies: (1) augment $\mathcal{D}_{\text{cal}}$ with FF_{-40} samples for hold-dominated designs; (2) calibrate on $N \geq 3$ chips from different production lots; (3) on Quartus targets, set $\text{NUM_PARALLEL_PROCESSORS} = 1$ for bit-exact audit-trail reproducibility.

I. Summary

The four experiments jointly show that the conformal envelope calibrated on tool reports transfers to silicon at 87% aggregate empirical coverage, with violations concentrated in three documented mechanisms. The framework's sign-off rule produced zero false-positive accepts, and every accepted design operated above its target frequency.

XI. CONCLUSION

We presented ROBIN-FPGA, a distributionally-robust RL framework for FPGA timing closure with a conformal sign-off envelope. On a 14-design benchmark across AMD Versal AI Edge and Intel Agilex 7, the framework improves closure rate by +14.0 pp over the strongest baseline, reduces inter-seed WNS variance by a median of $\sim 1.2\times$, and provides calibrated sign-off envelopes that track target coverage within ± 3 pp at $\alpha = 0.05$ (measured 92.7%) under exchangeable calibration. A two-stage curriculum enables cross-family transfer at 5% of the from-scratch compute. Hardware-in-the-loop measurements on 10 accepted designs across 5 physical chips and 8 thermal-chamber PVT corners yield 87% aggregate silicon-stage coverage, with violations traced to three documented failure modes and zero false-positive sign-offs. Future work includes joint timing–power optimisation, model-based variants to amortise P&R cost, and adaptive conformal methods for continuous tool evolution.

APPENDIX A: EXPERIMENTAL INFRASTRUCTURE AND REPRODUCIBILITY

All P&R runs were executed on a 96-core x86 cluster (dual AMD EPYC 9654, 512 GB RAM per node) under Vivado 2024.2 and Quartus Prime Pro 24.1. Each run is isolated in a dedicated process with a fixed random seed passed via `-seed` (Vivado) and `--seed` (Quartus). The per-run audit manifest records the tool version string, seed value, directive bundle hash, post-route WNS, TNS, utilization dictionary, and wall-clock runtime. Manifest files are archived as `.json.gz` alongside the bitstream and verified against the SHA-256 hash of the policy checkpoint at acceptance time. The full set of 4,900 run logs, Tcl scripts, and training checkpoints is released at the companion repository to enable bit-exact reproduction on matching tool versions. Learning curves, CVaR statistics,

and conformal quantiles are computed from these logs using the provided analysis scripts; no post-hoc smoothing beyond the 10/90-percentile band reported in Figure 9 is applied.

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